COMP9001
Lecture 3
Introduction to Programming
Control Flow: Conditionals

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Version 1.0

Inspired by Hazem El-Alfy 2024 slides

CRICOS 00026A TEQSA PRV12057

We recognise and pay respect to the Elders and communities – past, present, and emerging – of the lands that the University of Sydney's campuses stand on. For thousands of years they have shared and exchanged knowledges across innumerable generations for the benefit of all.

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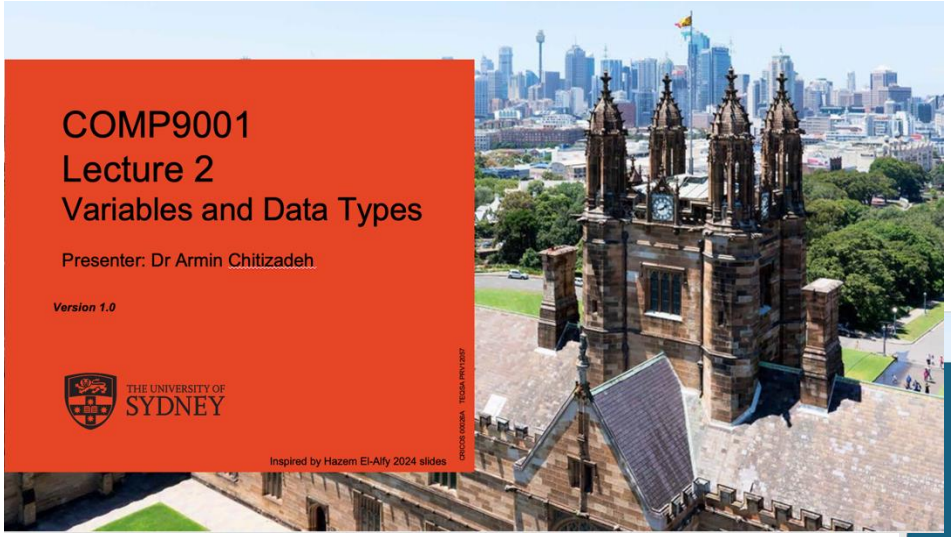
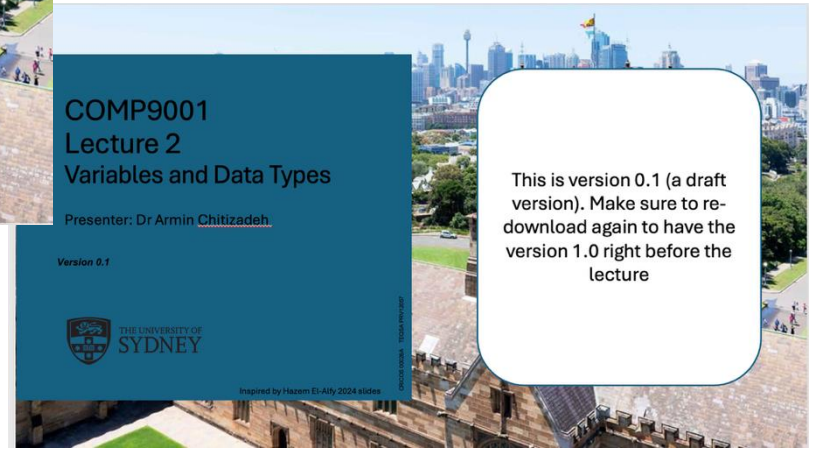
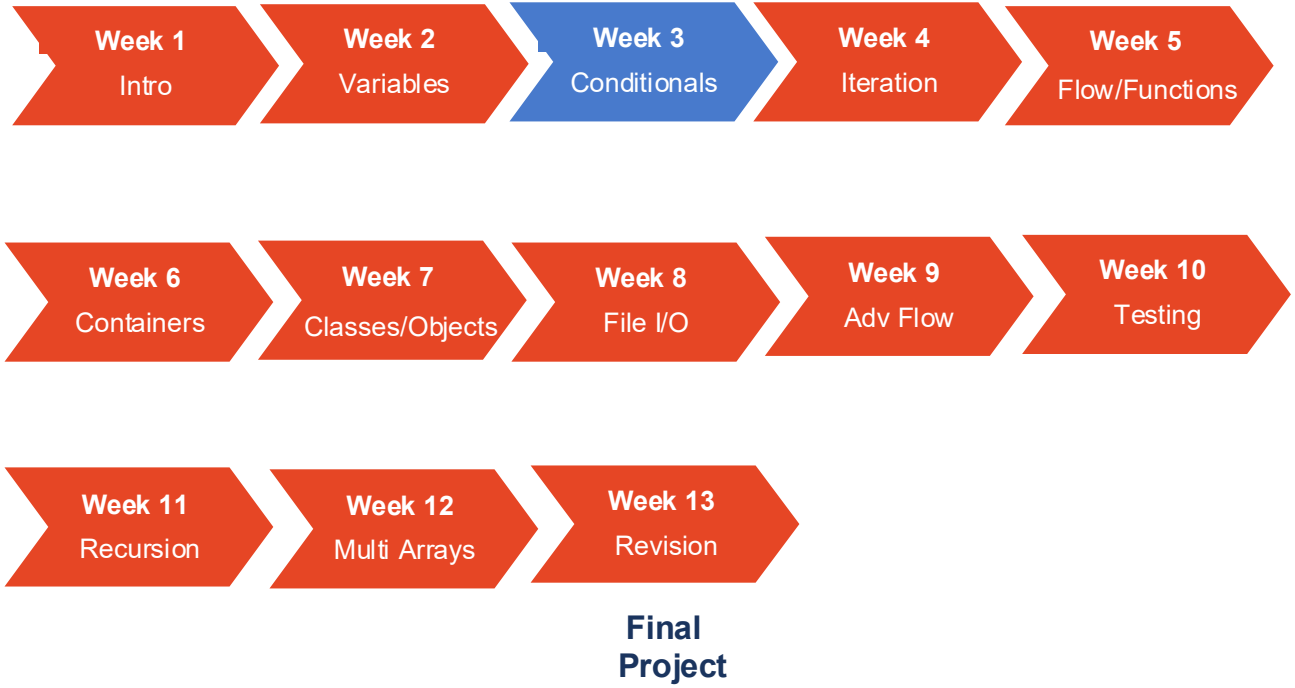
What is with Orange and Blue versions?



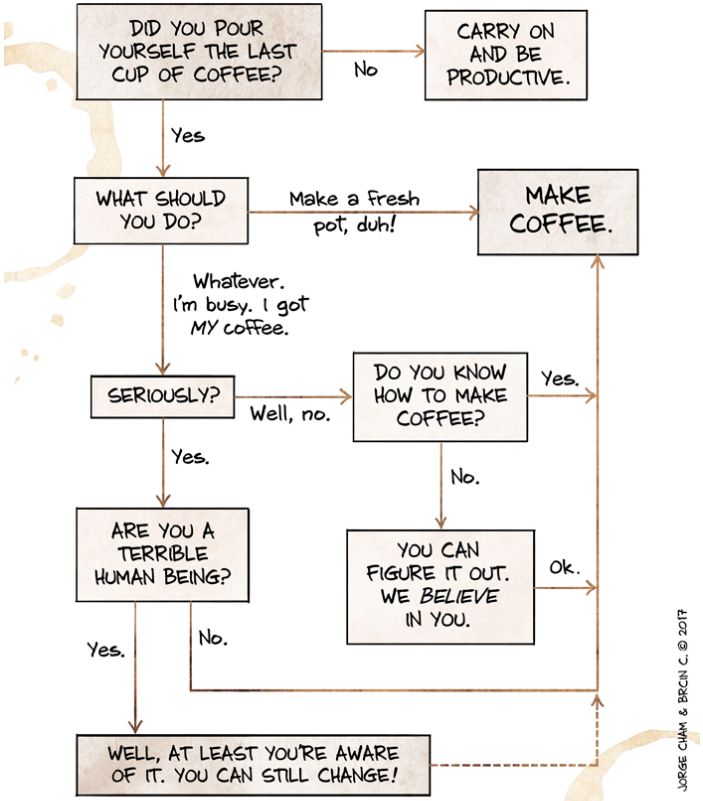


Where are we in the unit?



We make about 35,000 decisions each day. Tell us about one decision you have made very recently.

The Office Coffee Flowchart



WWW.PHDCOMICS.COM

First Quarter!

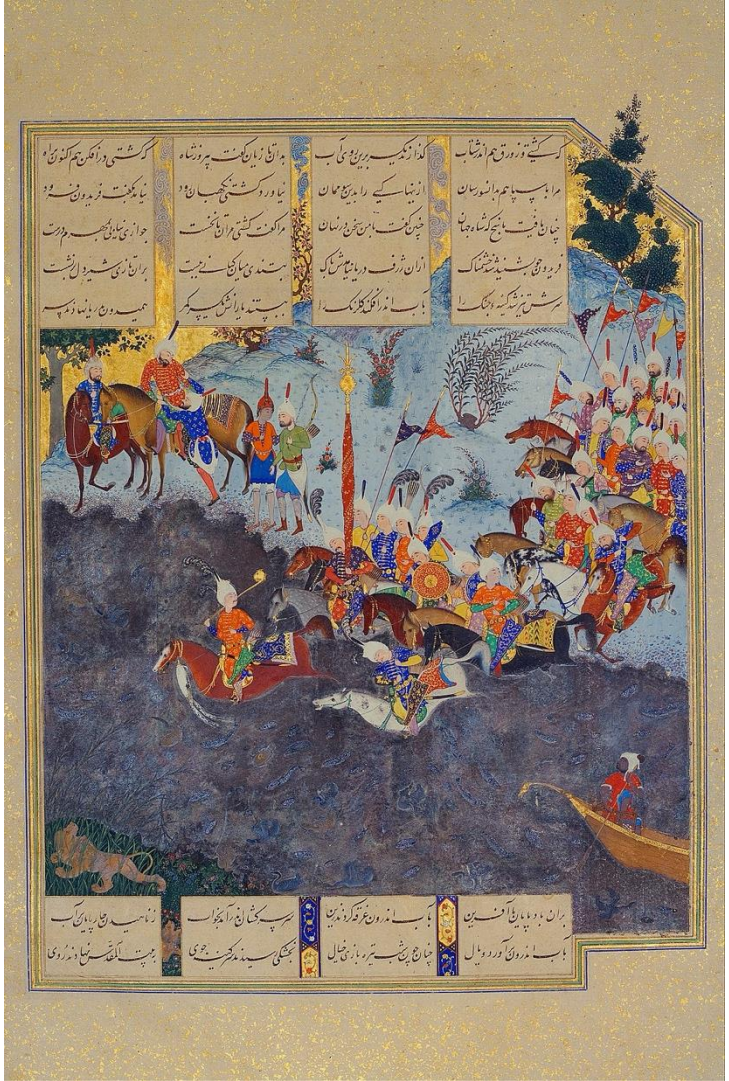
- Quarter-mark of the semester.

- Programming Basics
  - Input/Output
    - `input()`
    - `print()`
  - Built-in functions
  - Variables & Data Types
  - Formulas to Expressions
    - Operators
    - Order of operations

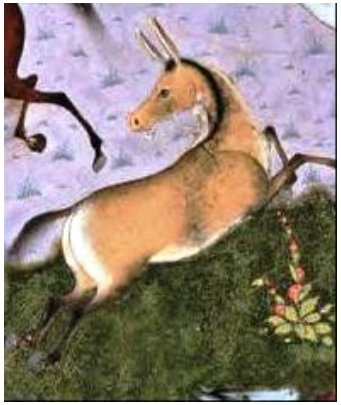
- Quarter-semester Survey on programming skills.

The Epic of Kings

Akvan Div



Faridun crosses the River Dijla



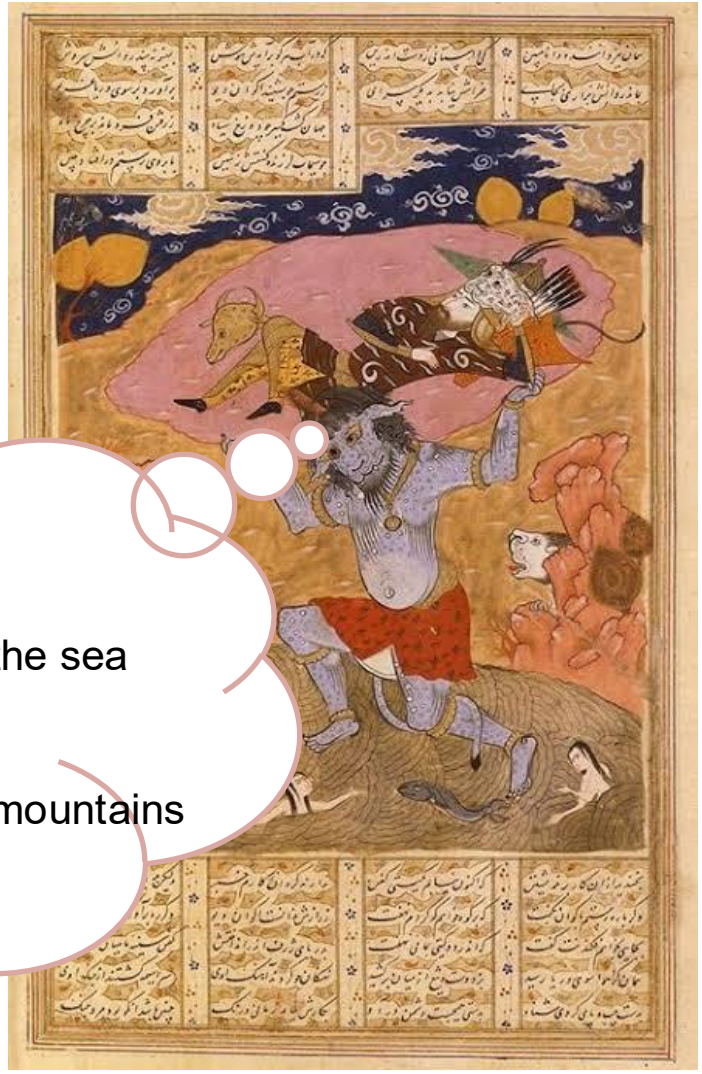
Akvan the beast asks:

Do you want to be thrown into

- the sea or
- mountains

If he answers “mountains”
Then I will throw him into the sea

If he answers “sea”,
Then I will throw him into mountains



Why do we need to make decisions in computer programs?

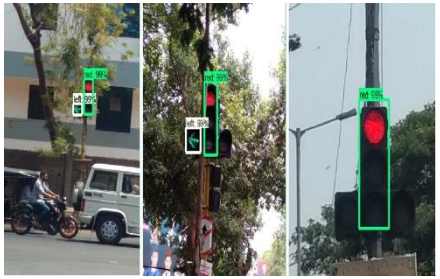
Teacher View

Student View

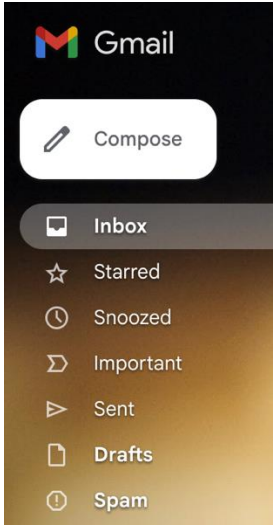
Why do we need to make decisions in real life?

Eligible orders over $65* will receive free delivery.

Traffic Light Detection and Recognition for Self Driving Cars Using Deep Learning



Spam email detections

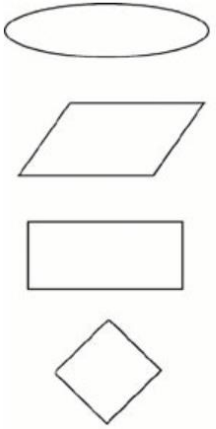


Why do we need to make decisions?

```python
# tailor4.py
customer_name = "Armin Chitizadeh"
customer_id = "3322445"
hips_size = 120
pant_size = 100
thigh = 28
pant = 139
total_fabric = ((hips_size + pant_size) * (thigh + pant))/10
FIXED_COST = 40
exchange_rate = input("Please give me the exchange rate Baba Yaga: ")

print("details of", customer_name, "with customer ID of", customer_id)
print("Customer Name:", customer_name)
print("Customer ID:", customer_id)
print("Hips: ", hips_size)
print("Pant Waist: ", pant_size)
print("Thigh: ", thigh)
print("Pant Length: ", pant)
print("Total fabric: ", total_fabric)

total_cost = int((40 + total_fabric/100) * int(exchange_rate))

print("Total cost: ", total_cost, "Toman")
```

Flowcharts

- Present plan of solution independent of programming language.



An oval indicates beginning or end of a program.

A parallelogram is a point where there is input to or ouput from the program.

A rectangle indicates the assignment of a value to a variable, constant, or parameter. the assigned value can be the reult of a computation. The computation would also be included in the rectangle.

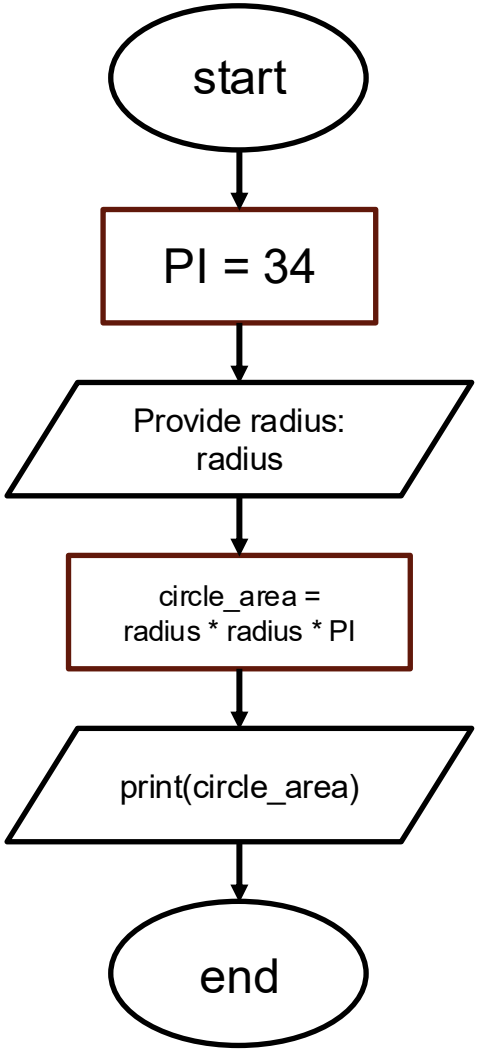
A diamond indicates a point where a decision is made.

- We will use these to visualize the flow of execution.

Image credit: https://www.owlnet.rice.edu/~ceng303/manuals/fortran/FOR3_3.html

Example

```
1 #Simple program
2 PI = 34
3 radius = int(input("Provide radius:"))
4 circle_area = radius * radius * PI
5 print(circle_area
```



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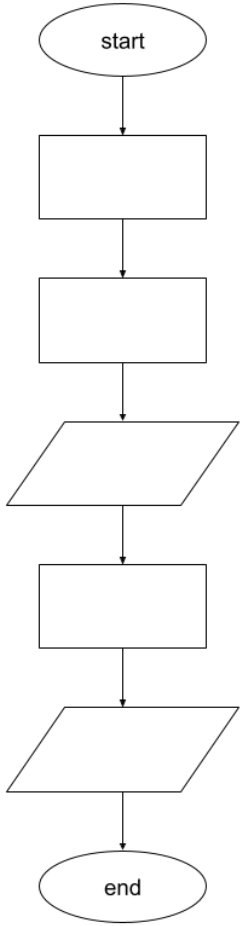


Making Decisions in Life

- We make decisions every day.
  - If happy, clap hands.
  - If **not** happy, stomp feet.
  - If I am thirsty **and** hungry, I will order a meal set.
  - If it is sunny **and** I'm feeling well, take a walk in the park.
    Else, stay home.
  - If I am tired **or** the time is between 10pm and 2am in the morning, I will sleep.
    Else, if it is Monday 6pm-10pm, I will work on INFO1110 labs.
    Else, if it is Tuesday 6pm-10pm, I will go kickboxing.
    On every other evening, I will farm on WoW with my mates!

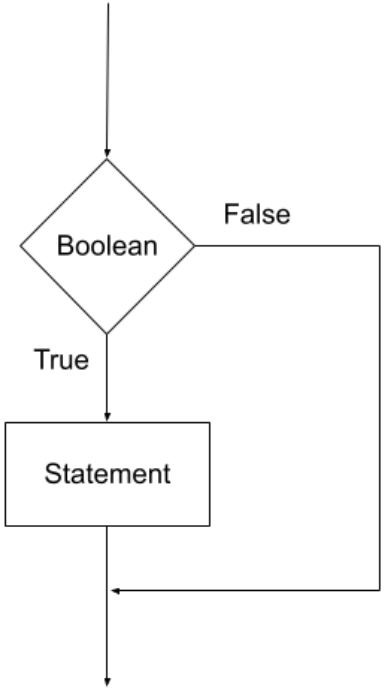
- Do you see any pattern?

Control Flow: if

- Syntax:

```
if <Boolean_expression>:
    Statement
```

- Flowchart



Recap: Boolean and Comparison Operator

- Boolean values: `True`, `False`
- A mixed type operator which takes operands of one type and returns another type (Boolean).

op	*meaning*	`True`	`False`
`==`	*equal*	`2 == 2`	`2 == 3`
`!=`	*not equal*	`3 != 2`	`2 != 2`
`<`	*less than*	`2 < 13`	`2 < 2`
`<=`	*less than or equal*	`2 <= 2`	`3 <= 2`
`>`	*greater than*	`13 > 2`	`2 > 13`
`>=`	*greater than or equal*	`3 >= 2`	`2 >= 3`

Comparisons with `int` operands and a `bool` result

How do you model Akvan the beast?

akvan.py

```python
# Akvan the beast model

rostam_answer = input("wakey wakey,\nwhere do you want me to throw you, sea or mountains? ")

if rostam_answer == "sea":
    print("I'm throwing you into MOUNTAINS! Ha Ha Ha!")

if rostam_answer == "mountains":
    print("I'm throwing you into the SEA! Ha Ha Ha!")
```

If he answers “mountains”
Then I will throw him into the sea

If he answers “sea”,
Then I will throw him into mountains

What about pass or fail?

- If you get 50 or above, you will pass.
- If you get less than 50, you will fail.

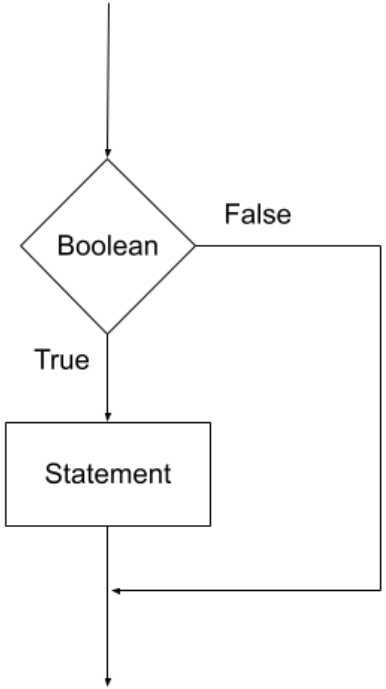
Ed link to pass or fail

Can we do better?

Can you be more efficient?

- If you get 50 or above, you will pass,
  - Otherwise, you will fail

if …



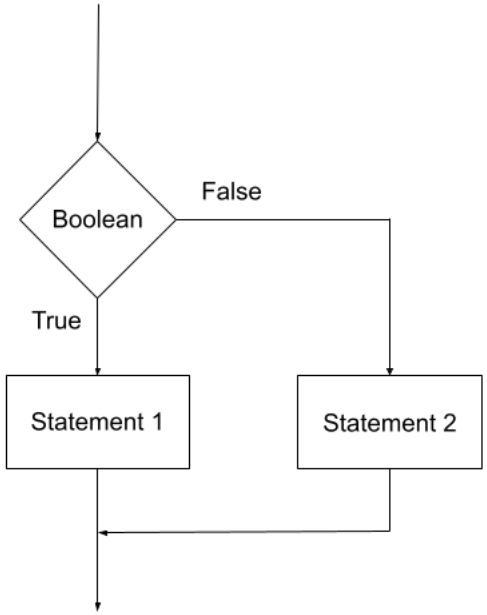
Control Flow: if… else

- Syntax:

```
if <Boolean expression>:
    Statement 1
else:
    Statement 2
```

Warning:
DO NOT put expressions after keyword `else`.

- Flowchart



Please rewrite the pass or fail python with if … else

- Ed link to pass or fail

How about multiple grades?

Result code	Result name	Mark range
HD	High distinction	85 - 100
DI	Distinction	75 - 84
CR	Credit	65 - 74
PS	Pass	50 - 64
FA	Fail	0 - 49

Can you break the beast below?

akvan.py

```python
# Akvan the beast model

rostam_answer = input("wakey wakey,\nwhere do you want me to throw you, sea or mountains? ")

if rostam_answer == "sea":
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if rostam_answer == "mountains":
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```

If he answers “mountains”
Then I will throw him into the sea

If he answers “sea”,
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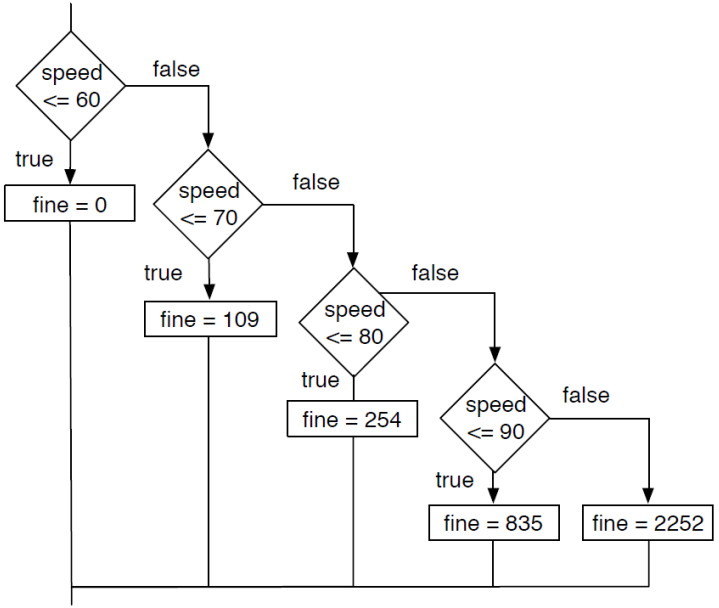
Control Flow: `if… elif…`

- `elif` abbreviation of else-if.
- When you have more than two possibilities and need more than two branches.
- No limit on the number of `elif` statements.
- If there is an `else` clause, it has to be at the end. Not syntactically needed to have one.
- Condition is checked in order of appearance. If the first is false, the next is checked and so on. Only the first true branch runs.

Example – How much `fine`?




```
if speed <= 60:
  fine = 0
elif speed <= 70:
  fine = 109
elif speed <= 80:
  fine = 254
elif speed <= 90:
  fine = 835
else:
  fine = 2252
```

Activity

Determine the value of fine if speed is:

- `-800`
- `90`
- `95`

If … elif

- Write a python file with if … elif to print the correct grade given the mark.

Link to ed

Result code	Result name	Mark range
HD	High distinction	85 - 100
DI	Distinction	75 - 84
CR	Credit	65 - 74
PS	Pass	50 - 64
FA	Fail	0 - 49

Break

What if you want to give an error if the given a value more than 100?

- We need more advanced condition (complex conditions)

Making Decisions in Life

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    Else, stay home.
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    On every other evening, I will farm on WoW with my mates!

- Do you see any pattern?

Making Decisions in Life

- If happy, clap hands.
- If **not** happy, stomp feet.

Condition: happy?	Action
True	
False	

Condition: not happy?	Action
True	
False	

Making Decisions in Life

- If I am thirsty **and** hungry, I will order a meal set.

thirsty	hungry	thirsty and hungry	Action
False	False		
False	True		
True	False		
True	True		

Making Decisions in Life

- If it is sunny **and** I'm feeling well, take a walk in the park. Else, stay home.

sunny?	feeling well?	sunny and feeling well?	Action
False	False		
False	True		
True	False		
True	True		

Recap: Boolean Operators

- Also called logical operators.

Operator	Operation	Example	Result
`not`	NOT	`not P`	True if P is False. False if P is True
`and`	AND	`P and Q`	True if P and Q are True. Otherwise, False.
`or`	OR	`P or Q`	True if either P or Q is True. Otherwise, False.

- Truth table presents result of every truth combination given inputs.

P	not P
True	False
False	True

P	Q	P and Q
False	False	False
False	True	False
True	False	False
True	True	True

P	Q	P or Q
False	False	False
False	True	True
True	False	True
True	True	True

Programs with Decisions

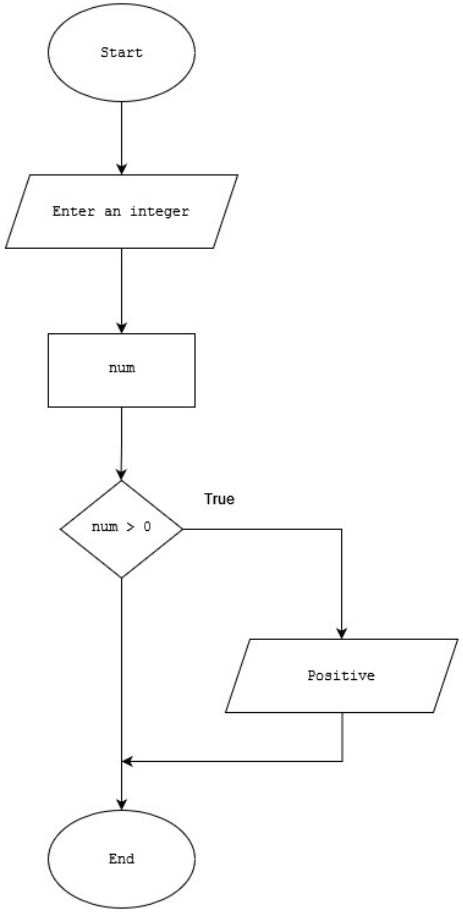
- Instruct computer to perform set of actions depending on whether a condition is fulfilled.

- Selection statements:
  - Conditional Execution – `if`
  - Alternative Execution – `if-else`
  - Chained conditionals – `if-elif`

Example – Simple Conditions

- Write a program that prompts user for an integer and display ‘positive’ if number is more than 0.

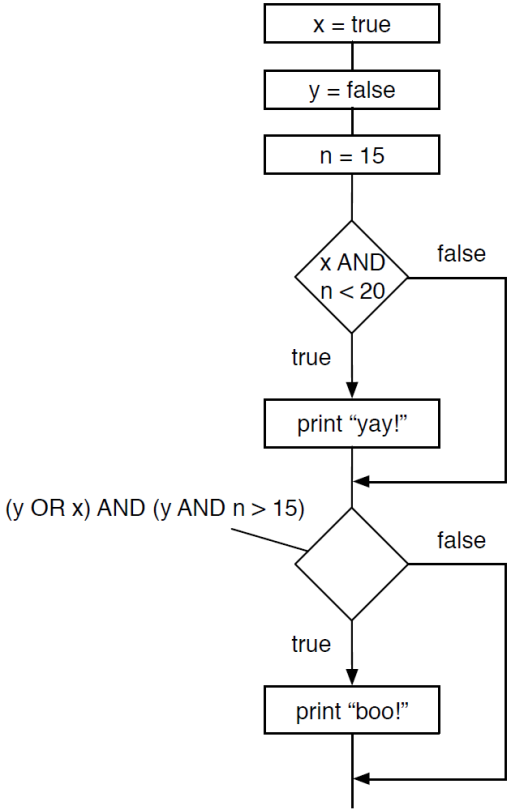
- Ed Lesson



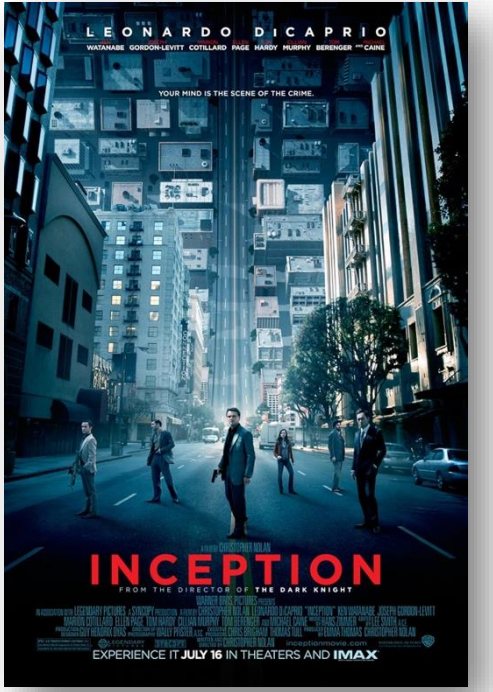
Example: Complex Condition

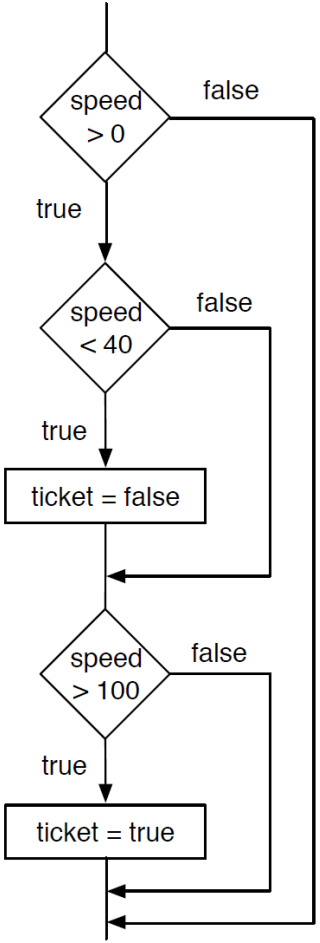
- What is the output of the program?

Conditions	Result
n<20	
x AND n<20	
y OR x	
y AND n>15	
(y OR x) AND (y AND n>15)	



Nested `if`


LEONARDO DICAPRIO
WATANABE GORDON-LEVITT COTILLARD PAGE HARDY MURPHY BERENGER CAINE
YOUR MIND IS THE SCENE OF THE CRIME.
INCEPTION
FROM THE DIRECTOR OF THE DARK KNIGHT
EXPERIENCE IT JULY 16 IN THEATERS AND IMAX


speed > 0
false
true
speed < 40
false
true
ticket = false
speed > 100
false
true
ticket = true

Example: Nested `if`

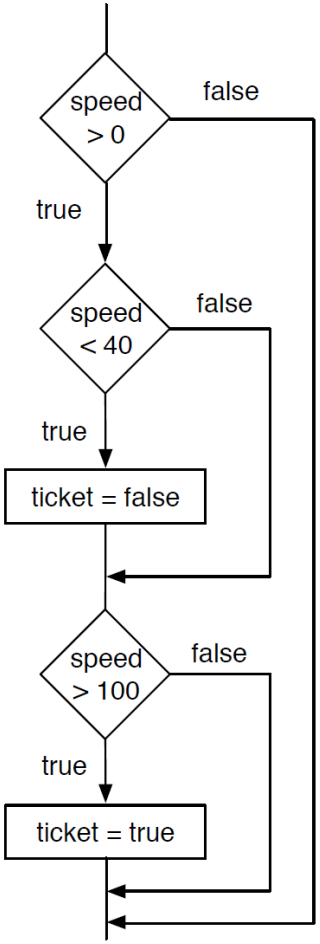
Activity

Predict the value in variable `ticket` at the end of the program if `speed` is initialized to:

- `5`
- `40`
- `80`
- `100`

Write a program to check your answer.

- Go to: Ed Lesson



Some good points:

- Remember to initialize *all* variables to a useful value.
- Safest to list out *all* possibilities condition can take when using `if`s.
- Not an issue, if it is a simple case. Being human, we can easily make a mistake and miss a case. :(
- `pass` statement is a null statement i.e. do nothing. Placeholder when statement is required syntactically but no codes needs to be executed.
- Remember when to use multiple ifs and when to use `if .. elif` s

positive2.py

```
1 print('Enter an integer:')
2 num = int(input())
3
4 if num > 0:
5     print('Positive')
6
7 if not(num > 0):
8     pass
```

Some good points:

When to use multiple `if`s vs `elif`s

Both are bad

bad_kylie.py

```python
# bad_kylie.py bad kylie minogue Grade program,

mark = int(input("Please enter the mark: "))
name = input("Please enter the name: ")

grade = "undefined"
if mark >= 85:
    grade = "HD"
if mark >= 75:
    grade = "DI"
if mark >= 65:
    grade = "CR"
if mark >= 50:
    grade = "PS"
if mark >= 0:
    grade = "FA"

title = ""
if name == "Kylie":
    title = "Her Majesty Queen "
else:
    title = ""

print(title, name, " recieved a grade of ", grade, sep="")
```

still_bad_kylie.py

```python
# still_bad_kylie.py bad kylie minogue Grade program,

mark = int(input("Please enter the mark: "))
name = input("Please enter the name: ")

grade = "undefined"
if mark >= 85 and mark < 100:
    grade = "HD"
if mark >= 75 and mark < 85:
    grade = "DI"
if mark >= 65 and mark < 75:
    grade = "CR"
if mark >= 50 and mark < 65:
    grade = "PS"
if mark >= 0 and mark < 50:
    grade = "FA"

title = ""
if name == "Kylie":
    title = "Her Majesty Queen "
else:
    title = ""

print(title, name, " recieved a grade of ", grade, sep="")
```

Some good points:

When to use multiple `if`s vs `elif`s

Good

kylie.py

```python
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name = input("Please enter the name: ")

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bad

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print(title, name, " recieved a grade of ", grade, sep="")
```

Can you write a better code to model Akvan the beast?

Akvan the beast asks:

Do you want to be thrown into

- the sea or
- mountains

If he answers “mountains”
Then I will throw him into the sea

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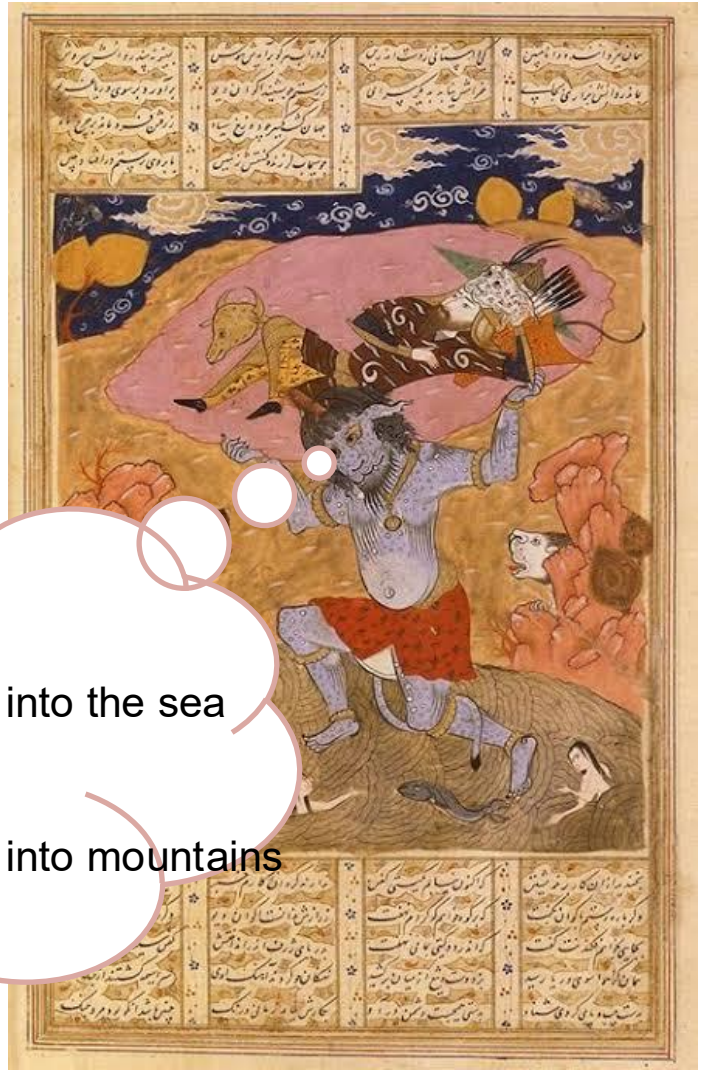
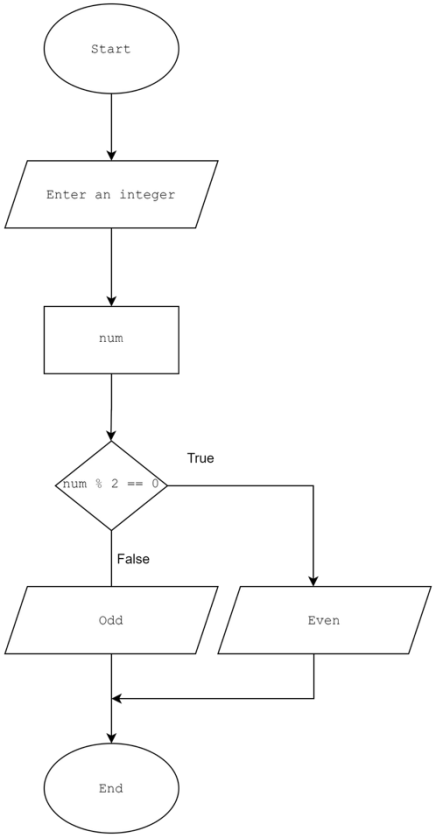


Exercises:

Example – Odd or Even

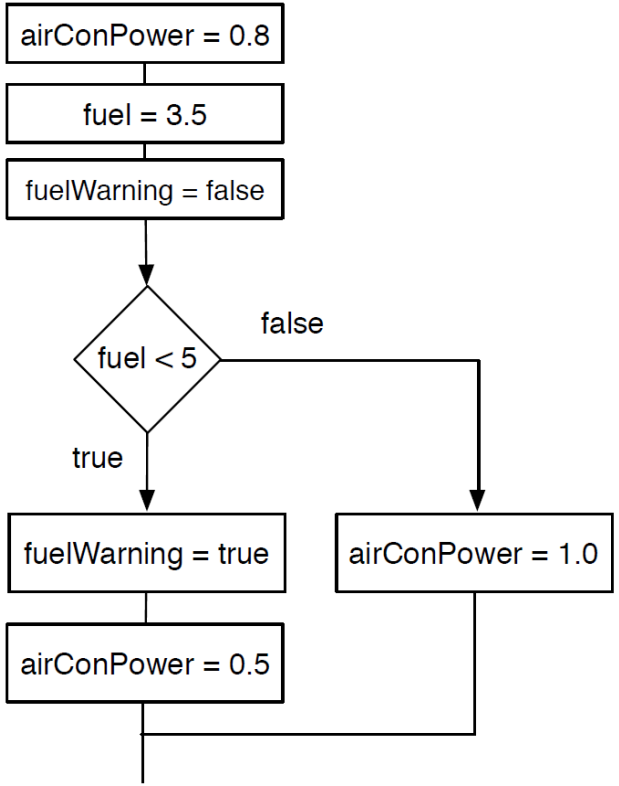
- Write a program that prompts user for an integer and display if it is odd or even.

- Ed Lesson



Example – Fuel Warning

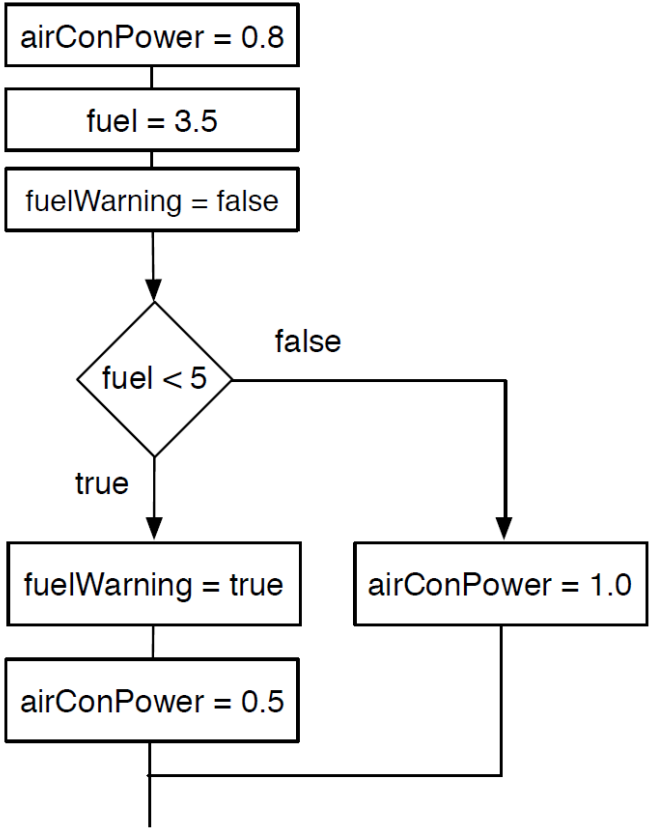
- What are the final values stored in `fuelWarning` and `airConPower`?
- Implement the flowchart in Python. https://edstem.org/au/courses/30774/lessons/98998/slides/679444
- Did we consider *all* possibilities of `fuel`?



Deck Check

- Track each variable value and how it changes.
- Test by picking values that are normal, abnormal, on the boundary…

fuel	airConPower	fuelWarning
0		
5		



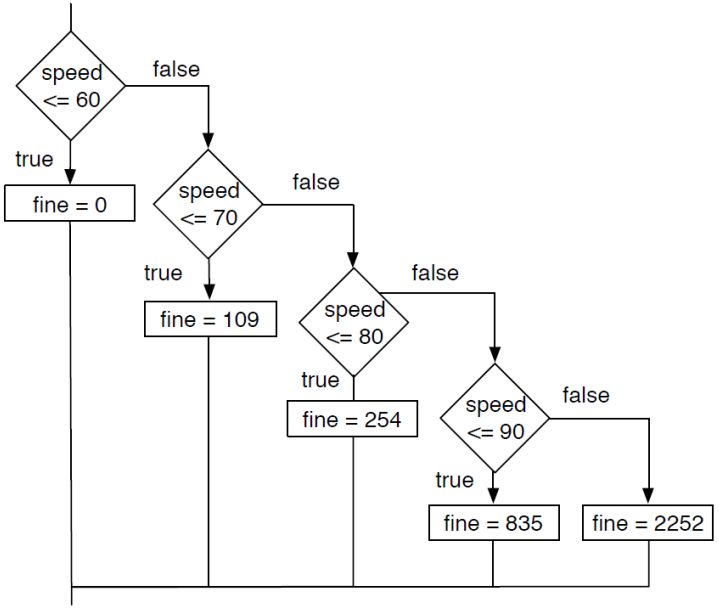
Example - Checking a, b, c

- What will be the output of this program?

```
1  char = 'a'
2
3  if char == 'a':
4      print('a')
5  elif char == 'b':
6      print('b')
7  elif char == 'c':
8      print('c')
9
10 print('Unknown')
11
```

- What happens if `char` is set to 'z'?
- Ed Lesson

Example – How much `fine`?




```
if speed <= 60:
  fine = 0
elif speed <= 70:
  fine = 109
elif speed <= 80:
  fine = 254
elif speed <= 90:
  fine = 835
else:
  fine = 2252
```

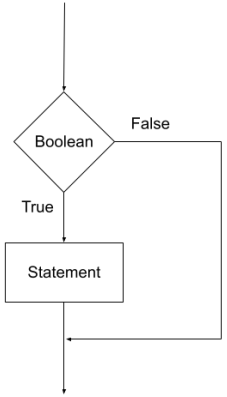
Activity

Determine the value of fine if speed is:

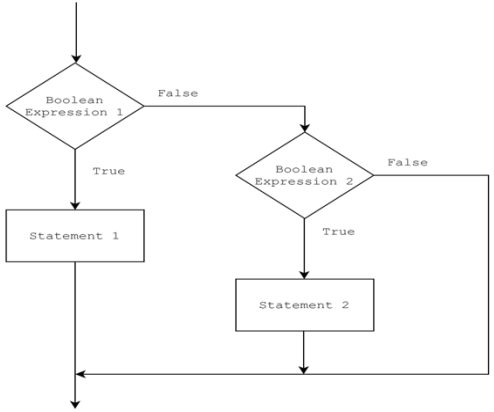
- `-800`
- `90`
- `95`

Summary

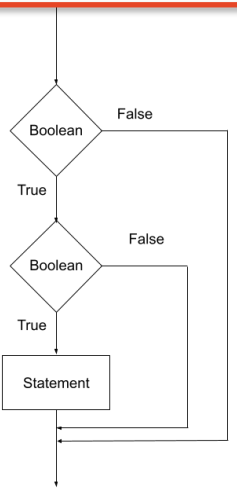
- Selection statements allows the flow of execution to follow a specific branch *depending on the condition.*
- Different types of branching:



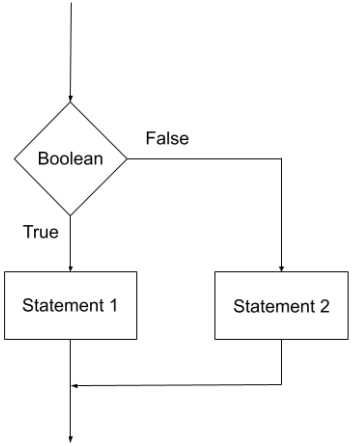
`if`



`if elif`



Nested `ifs`



`if-else`

Summary

- Selection statements allows the flow of execution to follow a specific branch *depending on the condition.*
- Different types of branching:

if.py

```python
# simple if
number = 11

if number > 10:
    result = number % 10

print(result)
```

if

```python
# simple if elif
number = 7

if number > 10:
    result = number % 10
elif number > 2:
    result = number % 2

print(result)
```

if elif

```python
#simple Nested if
number = 7

if number < 10:
    if number > 0:
        result = number * 10

print(result)
```

Nested ifs

```python
#simple if-else
number = 7

if number > 10:
    result = number % 10
else:
    result = number

print(result)
```

if-else

Summary

- How to turn flow chart into python code
- How to turn python code into flow chart

Menti!

https://www.menti.com/alvi33a2cge8

Reading for this week

- **Chapter 5. Conditionals.** Downey, A. B. (2015). Think Python: How to Think Like a Computer Scientist (2e ed.). O'Reilly Media, Incorporated.

- **Chapter 3.6 – 3.12.** Liang, Y. (2022). Introduction to Python Programming and Data Structures, Global Edition. Pearson Education Limited.

Updates